

Namespace SixLabors.ImageSharp

Classes

Configuration

Provides configuration which allows altering default behaviour or extending the library.

GeometryUtilities

Utility class for common geometric functions.

GraphicOptionsDefaultsExtensions

Adds extensions that allow the processing of images to the `Image<TPixel>` type.

GraphicsOptions

Provides configuration for controlling how graphics operations are rendered, including antialiasing, pixel blending, alpha composition, and coverage thresholding.

Image

Encapsulates an image, which consists of the pixel data for a graphics image and its attributes. For the non-generic `Image` type, the pixel type is only known at runtime. `Image` is always implemented by a pixel-specific `Image<TPixel>` instance.

ImageExtensions

Extension methods for the `Image` type.

ImageFormatException

The exception that is thrown when the library tries to load an image, which has format or content that is invalid or unsupported by ImageSharp.

ImageFrame

Represents a pixel-agnostic image frame containing all pixel data and `ImageFrameMetadata`. In case of animated formats like gif, it contains the single frame in a animation. In all other cases it is the only frame of the image.

ImageFrameCollection

Encapsulates a pixel-agnostic collection of `ImageFrame` instances that make up an `Image`.

ImageFrameCollectionExtensions

Extension methods for `ImageFrameCollection<TPixel>`.

ImageFrameCollection<TPixel>

Encapsulates a pixel-specific collection of `ImageFrame<TPixel>` instances that make up an `Image<TPixel>`.

ImageFrame<TPixel>

Represents a pixel-specific image frame containing all pixel data and **ImageFrameMetadata**. In case of animated formats like gif, it contains the single frame in a animation. In all other cases it is the only frame of the image.

ImageInfo

Contains information about the image including dimensions, pixel type information and additional metadata

ImageMetadataExtensions

Extension methods for the **ImageMetadata** and **ImageFrameMetadata** types.

ImageProcessingException

The exception that is thrown when an error occurs when applying a process to an image.

Image<TPixel>

Encapsulates an image, which consists of the pixel data for a graphics image and its attributes. For generic **Image<TPixel>**-s the pixel type is known at compile time.

IndexedImageFrame<TPixel>

A pixel-specific image frame where each pixel buffer value represents an index in a color palette.

InvalidImageContentException

The exception that is thrown when the library tries to load an image which contains invalid content.

Matrix3×2Extensions

Extension methods for the **Matrix3×2** struct.

UnknownImageFormatException

The exception that is thrown when the library tries to load an image which has an unknown format.

Structs

Color

Represents a color value that is convertible to any **IPixel<TSelf>** type.

ColorMatrix

A structure encapsulating a 5×4 matrix used for transforming the color and alpha components of an image.

DenseMatrix<T>

Represents a dense matrix with arbitrary elements. Components that are adjacent in a column of the matrix are adjacent in the storage array. The components are said to be stored in column major order.

Number

Represents an integral number.

PixelAccessor<TPixel>

Provides efficient access the pixel buffers of an **Image<TPixel>**.

Point

Represents an ordered pair of integer x- and y-coordinates that defines a point in a two-dimensional plane.

PointF

Represents an ordered pair of single precision floating point x- and y-coordinates that defines a point in a two-dimensional plane.

Rational

Represents a number that can be expressed as a fraction.

Rectangle

Stores a set of four integers that represent the location and size of a rectangle.

RectangleF

Stores a set of four single precision floating points that represent the location and size of a rectangle.

SignedRational

Represents a number that can be expressed as a fraction.

Size

Stores an ordered pair of integers, which specify a height and width.

SizeF

Stores an ordered pair of single precision floating points, which specify a height and width.

Interfaces

IDeepCloneable

An interface for objects that can be cloned. This creates a deep copy of the object.

IDeepCloneable<T>

A generic interface for a deeply cloneable type.

Enums

ByteOrder

The byte order of the data stream.

ColorHexFormat

Specifies the channel order when formatting or parsing a color as a hexadecimal string.

ReadOrigin

Specifies the position in a stream to use for reading.

Delegates

PixelAccessorAction<TPixel>

A delegate to be executed on a `PixelAccessor<TPixel>`.

PixelAccessorAction<TPixel1, TPixel2>

A delegate to be executed on two instances of `PixelAccessor<TPixel>`.

PixelAccessorAction<TPixel1, TPixel2, TPixel3>

A delegate to be executed on three instances of `PixelAccessor<TPixel>`.

Namespace SixLabors.ImageSharp.Drawing

Classes

ArcLineSegment

Represents a line segment that contains radii and angles that will be rendered as a elliptical arc.

ClipPathExtensions

Provides extension methods to **IPath** that allow the clipping of shapes.

ComplexPolygon

Represents a complex polygon made up of one or more shapes overlayed on each other, where overlaps causes holes.

CubicBezierLineSegment

Represents a line segment that contains a lists of control points that will be rendered as a cubic bezier curve

EllipsePolygon

An elliptical shape made up of a single path made up of one of more **ILineSegments**.

EmptyPath

A path that is always empty.

LinearGeometry

Represents retained linearized geometry that can be consumed directly by drawing backends.

LinearLineSegment

Represents a series of control points that will be joined by straight lines

OutlinePathExtensions

Extensions to **IPath** that allow the generation of outlines.

Path

A aggregate of **ILineSegments** making a single logical path.

PathBuilder

Allow you to derivatively build shapes and paths.

PathCollection

A aggregate of **IPaths** to apply common operations to them.

PathExtensions

Convenience methods that can be applied to shapes and paths.

PiePolygon

A pie sector polygon defined by a center point, radii, rotation, and arc sweep.

Polygon

A shape made up of a single closed path made up of one of more [ILineSegments](#)

RectanglePolygon

A closed rectangular path defined by four straight edges.

RegularPolygon

A shape made up of a single path made up of one of more [ILineSegments](#)

RoundedRectanglePolygon

A closed rectangular path with rounded corners.

SplitPathExtensions

Extensions to [IPath](#) for splitting paths into dash segments without performing stroke expansion.

StarPolygon

A star-shaped polygon defined by alternating inner and outer radii.

Structs

LinearContour

Describes a single contour within a [LinearGeometry](#).

LinearGeometryInfo

Describes geometry-wide metadata for a [LinearGeometry](#) instance.

LinearSegment

Represents one derived linear segment within a [LinearGeometry](#).

SegmentEnumerator

Enumerates the derived linear segments in a [LinearGeometry](#).

SegmentInfo

Returns metadata about the point along a path.

Interfaces

ILineSegment

Represents a simple path segment

IPath

Represents a logic path that can be drawn.

IPathCollection

Represents a logic path that can be drawn

ISimplePath

Represents a simple (non-composite) path defined by a series of points.

Enums

BooleanOperation

Specifies the type of boolean operation to perform on polygons.

IntersectionRule

Provides options for calculating intersection points.

LineCap

Specifies the shape to be used at the ends of open lines or paths when stroking.

LineJoin

Specifies how the connection between two consecutive line segments (a join) is rendered when stroking paths or polygons.

PathTypes

Describes the different type of paths.

Namespace SixLabors.ImageSharp.Web

Classes

ApplicationBuilderExtensions

Extension methods for [IApplicationBuilder](#) to simplify middleware registration.

CaseHandlingUriBuilder

Optimized helper methods for generating encoded Uris from URI components. Much of this code has been adapted from the MIT licensed .NET runtime.

ExifOrientationUtilities

Contains various helper methods for working with EXIF orientation values.

Format

Contains reusable static instances of known image formats.

FormatUtilities

Contains various helper methods for working with image formats based on the given configuration.

FormattedImage

A class encapsulating an image with a particular file encoding.

HMACUtilities

Provides methods to compute a Hash-based Message Authentication Code (HMAC).

ImageSharpBuilderExtensions

Extension methods for [ImageSharpBuilder](#) that allow configuration of services.

RequestAuthorizationUtilities

Contains various helper methods for authorizing image requests.

Resampler

Contains reusable static instances of known resampling algorithms.

ServiceCollectionExtensions

Extension methods for [IServiceCollection](#) to simplify middleware service registration.

Structs

FormatCommand

Represents a command for setting the type of format with which to encode processed images.

ImageCacheMetadata

Represents the metadata associated with an image file.

ImageMetadata

Represents the metadata associated with an image file.

ResamplerCommand

Represents a command for setting the type of sampler to use for resampling operations.

Interfaces

IImageSharpBuilder

Defines a contract for adding ImageSharp services.

Enums

CaseHandlingUriBuilder.CaseHandling

Provides Uri case handling options.

CommandHandling

Provides enumeration for handling **CommandCollection** instances when processing a request.

Namespace SixLabors.ImageSharp.Web. Azure.Resolvers

Classes

AzureBlobStorageCacheResolver

Provides means to manage image buffers within the [AzureBlobStorageCache](#).

AzureBlobStorageImageResolver

Provides means to manage image buffers within the Azure Blob file system.

Namespace SixLabors.ImageSharp.Web.AWS

Classes

AmazonS3BucketClient

Represents a scoped Amazon S3 client instance that is explicitly associated with a single S3 bucket. This wrapper provides a strongly-typed link between the client and the bucket it operates on, and optionally manages the lifetime of the underlying Amazon.S3.AmazonS3 Client.

Interfaces

IAWSS3BucketClientOptions

Provides a common interface for AWS S3 Bucket Client Options.

Namespace SixLabors.Fonts

Classes

Font

Defines a particular format for text, including font face, size, and style attributes. This class cannot be inherited.

FontCollection

Represents a collection of font families.

FontCollectionExtensions

Extension methods for [IFontCollection](#).

FontDescription

Provides basic descriptive metadata for the font.

FontException

Base class for exceptions thrown by this library.

FontFamilyNotFoundException

Exception for detailing missing font families.

FontGlyphMetrics

Represents a glyph metric from a particular font face.

FontMetrics

Represents a font face with metrics, which is a set of glyphs with a specific style (regular, italic, bold etc).

GlyphMissingException

Exception for detailing missing font families.

HorizontalMetrics

Represent the metrics of a font face specific to horizontal text.

InvalidFontFileException

Exception font loading can throw if it encounters invalid data during font loading.

InvalidFontTableException

Exception font loading can throw if it encounters invalid data during font loading.

KnownVariationAxes

Defines the registered design-variation axis tags for variable fonts. These tags are used with `FontVariation` to control font design axes. <https://learn.microsoft.com/en-us/typography/opentype/spec/dvaraxisreg>

LayoutModeExtensions

Extensions to `LayoutMode`.

LineLayout

Represents one laid-out line from a `TextBlock`.

LineLayoutEnumerator

Walks a `TextBlock` one laid-out line at a time.

MissingFontTableException

Exception font loading can throw if it finds a required table is missing during font loading.

SystemFonts

Provides a collection of fonts.

TextBlock

Represents text prepared for repeated line layout, measurement, and rendering.

TextMeasurer

Encapsulates logic for laying out and then measuring text properties.

TextMetrics

Encapsulates the full set of measurement results for laid-out text.

TextOptions

Provides configuration options for rendering and shaping text.

TextRun

Represents a run of text spanning a series of graphemes within a string.

VerticalMetrics

Represent the metrics of a font face specific to vertical text.

Structs

CaretPosition

Represents a caret line in laid-out text.

ClipQuad

Represents a rectangular clipping region as a convex quadrilateral. Allows for transformation by rotation, skew, or non-uniform scaling, resulting in non-axis-aligned edges.

FontFamily

Defines a group of type faces having a similar basic design and certain variations in styles.

FontRectangle

Stores a set of four single precision floating points that represent the location and size of a rectangle.

FontVariation

Represents a single variation axis setting for a variable font, consisting of a four-character tag and a value.

Glyph

Represents a font-specific glyph at the point size used for layout and rendering.

GlyphColor

Provides access to the color details for the current glyph.

GlyphMetrics

Represents one laid-out glyph entry in final layout order.

GraphemeMetrics

Represents one coalesced grapheme in final layout order.

LineMetrics

Encapsulates measured metrics for a single laid-out text line.

TextHit

Represents a hit-tested grapheme position in laid-out text.

TextPlaceholder

Represents an atomic inline placeholder with caller-supplied dimensions.

WordMetrics

Represents the positioned metrics for one Unicode word-boundary segment.

Interfaces

IFontCollection

A readable and writable collection of fonts.

IMetricsHeader

Defines the contract for the metrics header of a font face.

IReadOnlyFontCollection

Represents a readonly collection of fonts.

IReadOnlySystemFontCollection

Represents a readonly collection of Operating System fonts.

Enums

CaretMovement

Specifies a caret movement operation within laid-out text.

CaretPlacement

Specifies an absolute caret placement within a laid-out text scope.

ColorFontSupport

Specifies which color font formats are enabled for layout and rendering.

DecorationPositioningMode

Defines how text decorations (underline, overline, strikethrough) are positioned relative to font metrics.

FontStyle

The font styles

GlyphLayoutMode

Provides enumeration for the various layout mode of an individual glyph within a body of text.

GlyphType

Represents the various versions of a glyph records.

HintingMode

Defines modes to determine how to apply hinting. The use of mathematical instructions to adjust the display of an outline font so that it lines up with a rasterized grid.

HorizontalAlignment

Horizontal alignment modes.

KerningMode

Kerning is the contextual adjustment of inter-glyph spacing. This property controls metric kerning, kerning that utilizes adjustment data contained in the font.

LayoutMode

Defines modes to determine the layout direction of text.

TextAlignment

Text alignment modes.

TextAttributes

Provides enumeration of various text attributes.

TextBidiMode

Specifies how bidirectional text is resolved.

TextDecorations

Provides enumeration of various text decorations.

TextDirection

Specifies the writing direction for text.

TextEllipsis

Specifies ellipsis behavior when laid-out text is limited to a maximum number of lines.

TextHyphenation

Specifies hyphenation marker behavior when text breaks at hyphenation opportunities.

TextInteractionMode

Specifies how text interaction positions are modeled for laid-out text.

TextJustification

Text justification modes.

TextPlaceholderAlignment

Specifies how an inline placeholder is aligned against surrounding text.

VerticalAlignment

Vertical alignment modes.

WordBreaking

Defines modes to determine when line breaks should appear when words overflow their content box.

Namespace SixLabors.PolygonClipper

Classes

Contour

Represents a single polygon ring (outer contour or hole).

Polygon

Represents a complex polygon.

PolygonClipper

Performs boolean operations on polygons.

PolygonStroker

Generates polygonal stroke geometry for contours with configurable joins and caps.

StrokeOptions

Provides configuration options for geometric stroke generation.

Structs

Box2

Represents a bounding box.

Vertex

Represents a two-dimensional vertex with X and Y coordinates.

Enums

BooleanOperation

Specifies the type of boolean operation to perform on polygons.

LineCap

Specifies the shape to be used at the ends of open lines or paths when stroking.

LineJoin

Specifies how the connection between two consecutive line segments (a join) is rendered when stroking paths or polygons.

ResultTransition

Represents the result transition for a sweep event.